

Identidades trigonométricas

$$1) \quad 1 + \cot \alpha = \sin \alpha + \cos \alpha$$

$$2) \quad \sec \alpha - \cos \alpha = \tan \alpha \cdot \sin \alpha$$

$$\frac{1}{\cos \alpha} - \cos \alpha =$$

$$\frac{1 - \cos^2 \alpha}{\cos \alpha} =$$

$$\frac{\sin^2 \alpha}{\cos \alpha} =$$

$$= \tan \alpha \sin \alpha$$

$$3) \quad (\sin \alpha - \cos \alpha)(\csc \alpha + \sec \alpha) = \tan \alpha - \cot \alpha$$

$$= (\sin \alpha - \cos \alpha) \left(\frac{1}{\sin \alpha} + \frac{1}{\cos \alpha} \right)$$

$$= (\sin \alpha - \cos \alpha) \frac{\cos \alpha + \sin \alpha}{\sin \alpha \cos \alpha}$$

$$\frac{\sin \alpha \cos \alpha + \sin^2 \alpha - \cos^2 \alpha - \sin \alpha \cos \alpha}{\sin \alpha \cos \alpha}$$

$$= \frac{\sin^2 \alpha}{\sin \alpha \cos \alpha} - \frac{\cos^2 \alpha}{\sin \alpha \cos \alpha}$$

$$= \frac{\sin \alpha}{\cos \alpha} - \frac{\cos \alpha}{\sin \alpha}$$

$$= \tan \alpha - \cot \alpha$$

$$4) \quad 1 + \cot^2 \alpha = \csc^2 \alpha$$

$$= 1 + \frac{\cos^2 \alpha}{\sin^2 \alpha}$$

$$= \frac{\sin^2 \alpha + \cos^2 \alpha}{\sin^2 \alpha}$$

$$= \frac{1}{\sin^2 \alpha}$$

$$= \csc^2 \alpha$$

$$5) \quad (\sin \alpha + \cos \alpha)^2 = 2 \sin \alpha \cos \alpha + 1$$

$$\sin^2 \alpha + 2 \sin \alpha \cos \alpha + \cos^2 \alpha$$

$$= 2 \sin \alpha \cos \alpha + 1$$

$$6) \quad \frac{1 + \cos \alpha}{\sin \alpha} + \tan \alpha = \frac{1 + \cos \alpha}{\sin \alpha \cos \alpha}$$

$$\begin{aligned}
 &= \frac{1 + \cos \alpha}{\sin \alpha} + \frac{\sin \alpha}{\cos \alpha} = \\
 &= \frac{\cos \alpha + \cos^2 \alpha + \sin^2 \alpha}{\sin \alpha \cos \alpha} = \\
 &= \frac{1 + \cos \alpha}{\sin \alpha \cos \alpha}
 \end{aligned}$$

$$7) \quad \frac{1 + \tan \alpha}{1 + \cot \alpha} = \tan \alpha$$

$$\begin{aligned}
 &= \frac{1 + \tan \alpha}{1 + \frac{1}{\tan \alpha}} \\
 &= \frac{1 + \tan \alpha}{\frac{\tan \alpha + 1}{\tan \alpha}} \\
 &= \tan \alpha
 \end{aligned}$$

$$8) \quad \frac{1}{\frac{1}{\cos^2 \alpha} (1 - \sin^2 \alpha)} + \tan^2 \alpha = \sec^2 \alpha$$

$$\begin{aligned}
 &= \frac{1}{\frac{1}{\cos^2 \alpha} \cos^2 \alpha} + \tan^2 \alpha \\
 &= 1 + \tan^2 \alpha \\
 &= \sec^2 \alpha
 \end{aligned}$$

$$9) \quad (\sec^2 \alpha - 1) \csc^2 \alpha \cot \alpha = \frac{1}{\sin \alpha \cos \alpha}$$

$$\begin{aligned}
 &= \tan^2 \alpha \frac{1}{\sin^2 \alpha} \frac{\cos \alpha}{\sin \alpha} \\
 &= \frac{\sin^2 \alpha}{\cos^2 \alpha} \frac{1}{\sin^2 \alpha} \frac{\cos \alpha}{\sin \alpha} = \\
 &= \frac{1}{\sin \alpha \cos \alpha}
 \end{aligned}$$

$$10) \quad (\sin \alpha + \cos \alpha)^2 - \tan \alpha \cos^2 \alpha = 1 + \sin \alpha \cos \alpha$$

$$\begin{aligned}
 &= \sin^2 \alpha + 2 \sin \alpha \cos \alpha + \cos^2 \alpha - \frac{\sin \alpha}{\cos \alpha} \cos^2 \alpha \\
 &= 1 + 2 \sin \alpha \cos \alpha - \sin \alpha \cos \alpha \\
 &= 1 + \sin \alpha \cos \alpha
 \end{aligned}$$

$$11) \quad (1 + \sin \alpha) \tan \alpha = \sec \alpha (\sec \alpha + 1 - \cos^2 \alpha)$$

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$$12) \quad \tan \alpha + \cot \alpha = \frac{1}{\sin \alpha \cos \alpha}$$

$$= \frac{\sin^2 \alpha + \cos^2 \alpha}{\sin \alpha \cos \alpha}$$

$$= \frac{1}{\sin \alpha \cos \alpha}$$

$$13) \quad (\sin \alpha + \cos \alpha)^2 + (\cos \alpha - \sin \alpha)^2 = 2$$

$$= \sin^2 \alpha + 2 \sin \alpha \cos \alpha + \cos^2 \alpha + \cos^2 \alpha - 2 \sin \alpha \cos \alpha + \sin^2 \alpha =$$

$$= 2\sin^2 \alpha + 2\cos^2 \alpha$$

$$= 2(\sin^2 \alpha + \cos^2 \alpha)$$

$$= 2$$

$$14) \quad \frac{(1+\cos \alpha)(1-\cos \alpha)}{\cos \alpha} = \sec \alpha - \cos \alpha$$

$$= \frac{1 - \cos \alpha + \cos \alpha - \cos^2 \alpha}{\cos \alpha}$$

$$= \frac{1 - \cos^2 \alpha}{\cos \alpha}$$

$$= \frac{1}{\cos \alpha} - \frac{\cos^2 \alpha}{\cos \alpha}$$

$$= \sec \alpha - \cos \alpha$$

$$15) \quad \frac{1+\cot \alpha}{\csc \alpha} = \sin \alpha \cos \alpha$$

$$= \frac{1 + \frac{\cos \alpha}{\sin \alpha}}{\frac{1}{\sin \alpha}} =$$

$$= \sin \alpha + \frac{\sin \alpha \cos \alpha}{\sin \alpha} =$$

$$= \sin \alpha \cos \alpha$$

$$16) \quad (\sin \beta + \cos \beta)^2 - \tan \beta \cos \beta = 1 + \sin \beta \cos \beta$$

$$= \sin^2 \beta + 2 \sin \beta \cos \beta + \cos^2 \beta - \frac{\sin \beta}{\cos \beta} \frac{\cos^2 \beta}{1}$$

$$= 1 + 2 \sin \beta \cos \beta - \sin \beta \cos \beta$$

$$1 + \sin \beta \cos \beta$$

Verdadero o Falso

$$17) \quad \cot \alpha = \frac{\cos \alpha}{\sin \alpha} \quad \text{V}$$

$$18) \quad (\sin \alpha + \cos \alpha)^2 = \sin^2 \alpha + \cos^2 \alpha \quad \text{F}$$

$$\sin^2 \alpha + 2 \sin \alpha \cos \alpha + \cos^2 \alpha$$

$$2 \sin \alpha \cos \alpha + 1$$

$$19) \quad \tan \epsilon \cot \epsilon = 1$$

$$\tan \epsilon \frac{1}{\tan \epsilon} = 1 \quad \text{V}$$

$$20) \quad \tan \alpha + \cot \alpha = \frac{1}{\sin \alpha + \cos \alpha} \quad \text{V}$$

$$\frac{\sin \alpha}{\cos \alpha} + \frac{\cos \alpha}{\sin \alpha} =$$

$$\frac{\sin^2 \alpha + \cos^2 \alpha}{\sin \alpha \cos \alpha}$$

$$\frac{1}{\sin \alpha \cos \alpha}$$

Hallar el valor de todas las funciones trigonométricas:

$$21) \quad \tan \theta = \frac{3}{4}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

$$\cot \theta = \frac{1}{\frac{3}{4}}$$

$$\cot \theta = \frac{4}{3}$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$\sec^2 \theta = \tan^2 \theta + 1$$

$$\sec \theta = \sqrt{\tan^2 \theta + 1}$$

$$\sec \theta = \sqrt{\left(\frac{3}{4}\right)^2 + 1}$$

$$\sec \theta = \sqrt{\frac{9}{16} + 1}$$

$$\sec \theta = \sqrt{\frac{9 + 16}{16}}$$

$$\sec \theta = \sqrt{\frac{25}{16}}$$

$$\sec \theta = \frac{5}{4}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\cos \theta = \frac{1}{\frac{5}{4}}$$

$$\cos \theta = \frac{4}{5}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\sin \theta = \tan \theta \cos \theta$$

$$\sin \theta = \frac{3}{4} \cdot \frac{4}{5}$$

$$\sin \theta = \frac{3}{5}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{1}{\frac{3}{5}}$$

$$\csc \theta = \frac{5}{3}$$

